



Rentacar TG bot - roadmap

Backend: Django (DRF)

Bot: Aiogram

DB: PostgreSQL

Infra: Docker (+ Nginch xitoyiriy)

1) Project setup

1.1 Papka struktura (repo)

```
rentacar/ ├── backend/ ├── bot/ ├── docker/ ├── scripts/ ├── .env ├──  
.env.example ├── .gitignore ├── docker-compose.yml └── README.md
```

Izoh:

- `.env` commit qilinmaydi, shuning uchun `.env.example` qo'shiladi.
 - `scripts/` (yoki Makefile) — run/migrate/format kabi buyruqlar uchun.
-

2) Backend (Django + DRF)

2.1 Django apps

- `accounts` — user / customer profili (tg_id bilan bog'lash)
- `cars` — mashinalar (narx, mavjudlik, rasmlar, status)
- `bookings` — bronlar (start/end, status, total_price)
- `common` — utils, base models, mixins, constants
- (ixtiyoriy) `notifications` — botga xabar yuborish, admin alert

- (ixtiyoriy) `payments` — to'lovlar bo'lsa

2.2 Booking logika (kritik qoidalar)

- Sanalar: `start_date < end_date`
- O'tgan sana: `start_date` bugundan oldin bo'lmasin
- Overlap: bitta mashinaga bir vaqt oralig'ida 2 ta booking bo'lmasin
- Statuslar (misol):
 - `pending` → `confirmed` → `finished`
 - `canceled` (har bosqichda mumkin)

2.3 API (minimal + kerakli endpointlar)

Cars

- `GET /api/cars/`
- `GET /api/cars/<id>/`
- `GET /api/cars/available/?start=YYYY-MM-DD&end=YYYY-MM-DD`

Bookings

- `POST /api/bookings/` (create)
- `GET /api/bookings/user/<tg_id>/` (user bookings)
- `PATCH /api/bookings/<id>/` (status/date update)
- `DELETE /api/bookings/<id>/` (cancel)

2.4 Auth (bot bilan ishlash)

Variantlardan biri tanlanadi:

- API key (eng sodda): bot request headerida `X-API-KEY`
- yoki JWT / service account (kattaroq loyihada)

3) Telegram Bot (Aiogram)

3.1 Bot strukturasi

```
bot/ ├── handlers/ ├── keyboards/ ├── middlewares/ ├── services/ ├──  
states/ ├── utils/ ├── config.py └── main.py
```

3.2 Asosiy flow

`/start`

→ Mashinalar 🚗

→ Mening bronlarim 📅

3.3 Booking flow (detal)

- 1) Car tanlash
- 2) Start date kiritish
- 3) End date kiritish
- 4) Summary ko'rsatish (car + sana + narx)
- 5) Confirm / Cancel

Validatsiyalar:

- start/end formatini tekshirish
 - start < end
 - availability (backend orqali)
-

4) Docker / Deploy (qisqa)

- `docker-compose.yml` : backend + db (+ bot)
 - `docker/` : Dockerfile'lar, entrypoint, nginx config (agar kerak bo'lsa)
-

5) Keyingi bosqich (MVP checklist)

- ☐ Cars CRUD (admin)
 - ☐ Availability check
 - ☐ Booking create + overlap protection
 - ☐ Booking list (tg_id bo'yicha)
 - ☐ Booking cancel/confirm
 - ☐ Bot states + keyboards
 - ☐ Logging + basic monitoring
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